Friction changes when slipping begins

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Static friction / ES: fricción estática / PT: atrito estático

Adjusts to prevent slip, up to a limit.

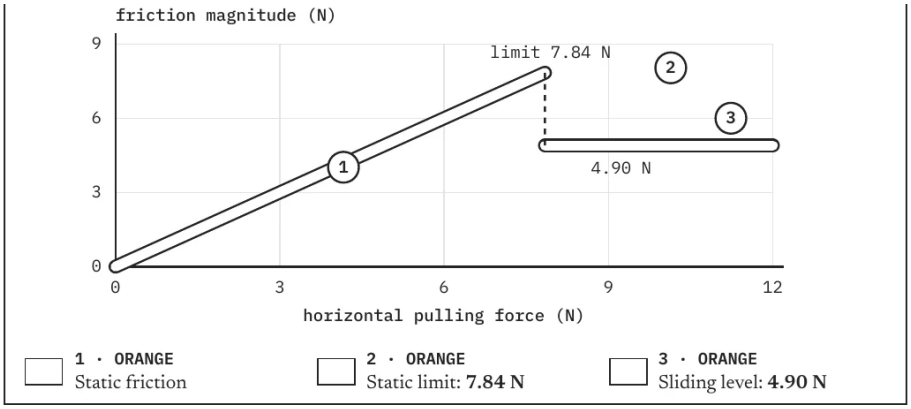
Also on your note page: Limit, Kinetic friction, Normal force.

B READ THE RELATIONSHIP

Static friction adjusts up to a limit; sliding friction acts during slip

As you read, find: at rest · just before slip · sliding.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

For this box, compare a 6 N pull while at rest with a 10 N pull after sliding starts. Use g = 9.8 m/s² ; pulls are horizontal.

Given N = 2(9.8) = 19.6 N μ s = 0.40 μ k = 0.25

Rule Check μ s N . Sliding: use μ k N , then net force.

Work f s ,max = 7.84 N at rest: f s = 6 N a = (10 – 4.9)/2

Answer 0; 2.55 m/s² At rest, then sliding.

E SAVE FOR THE FINAL MEETING

Surface friction is not a force that continually brakes XJ in empty space. Gravitational forces can still change its velocity, and atmospheric drag would matter in an atmosphere.

Read the physics: Static friction

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Push gently on a crate. If it stays still, static friction can balance the pull. A harder push needs more static friction, up to a maximum.

02 HOW IT WORKS

After slipping begins, use kinetic friction. In this model its size is the kinetic coefficient times the normal force. It opposes relative slipping at the contact.

03 WHAT TO CHECK

On a force trace, find the peak just before slip. Compare it with the steady-speed pull while the surfaces slide. Do not use the maximum static force for every pull.

04 TRY THE EXAMPLE FROM YOUR NOTES

For this box, compare a 6 N pull while at rest with a 10 N pull after sliding starts. Use g = 9.8 m/s² ; pulls are horizontal. 0; 2.55 m/s² At rest, then sliding.

Model check: Static friction adjusts up to a limit; sliding friction acts during slip.

05 CONNECT TO XJ

Surface friction is not a force that continually brakes XJ in empty space. Gravitational forces can still change its velocity, and atmospheric drag would matter in an atmosphere.

SAY IT BEFORE YOU WRITE

When do you switch from static to kinetic friction? Before sliding, friction ___. During sliding, we use ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Static friction / ES: fricción estática / PT: atrito estático

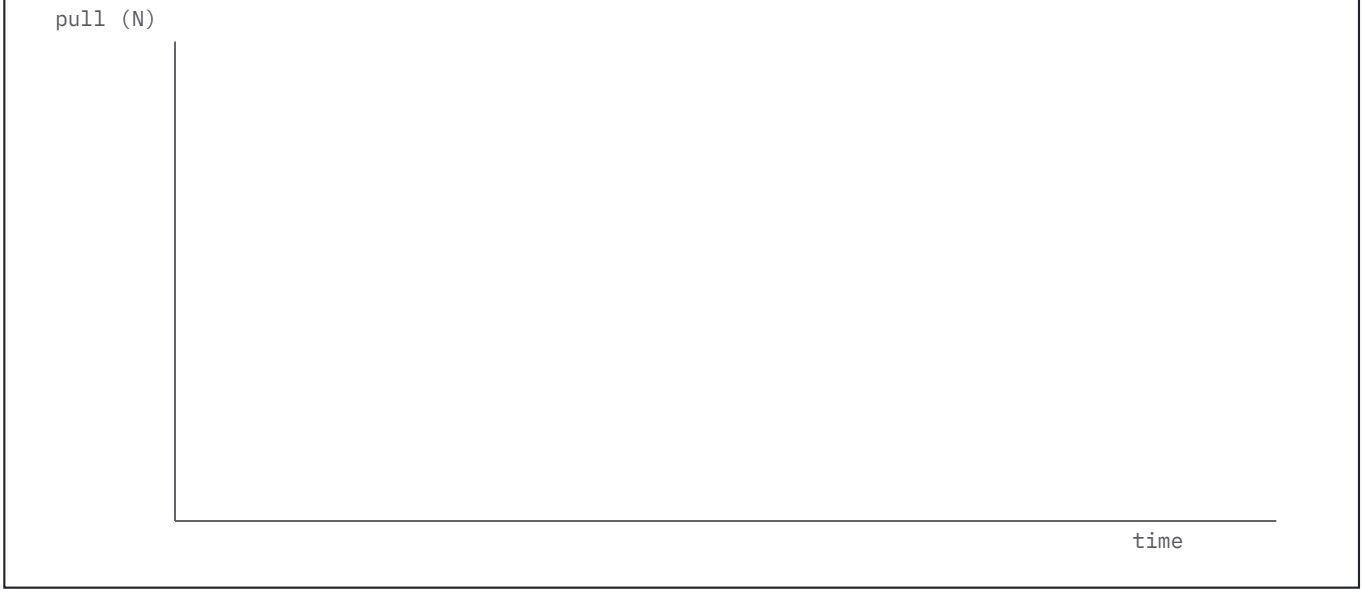
Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, static friction means …	You push a crate sideways and it stays still. What is happening where the surfaces touch? The surfaces stay ___. Friction acts toward ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Sketch a force trace. Label the peak just before slip and the steady-speed pull while the surfaces slide.

Labels you can use: at rest · just before slip · sliding.



Tell a partner: Before sliding, friction ___. During sliding, we use ___.

On assignment: Static friction

SECTION E / BACK / FINAL MEETING ONLY Name _____

A show cuts from a block sliding on a rough table to XJ coasting in empty space. The proposed caption says "Both will stop from surface friction."

PICK A JOB / DELIVER ONE PRODUCT

☐ **Effects engineer**

Specify the friction setting for each scene. Mark the contact surfaces, then tell the animator which motion needs another force to change.

☐ **Science broadcaster**

Record a correction for the cut. Your producer points to the surfaces in each scene and asks what could still bend XJ's path.

Word bank: at rest · just before slip · sliding.

Try: "Our desk recommends ___. This mark shows ___."

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

"We first said ___. Now we ___ because ___."

Editor's check: Explain how entering an atmosphere would change the force model.